

Special Topic: Enabling Techniques and Cutting-Edge Applications of Foundation Models

LEDNet: a multimodal foundation model for robust deepfake detection

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Abstract The proliferation of artificial intelligence (AI)-generated content on social media has escalated concerns regarding misinformation, privacy, and security. Current deepfake detection methods struggle to generalize effectively, failing to keep pace with evolving generative models like generative adversarial networks (GANs) and diffusion models. Inspired by the robustness of “real” and “fake” descriptions in the language world, in this paper, we introduce the language-enhanced deepfake detection network (LEDNet), a multimodal foundation model that leverages both linguistic and visual information to improve detection accuracy and adaptability. We are the first to exploit multi-modal technologies for generalized deepfake detection through an un-trained-backbone mechanism, which utilizes pre-trained vision-language models without the need for additional retraining. The model is powered by two key components: language-guided knowledge aggregation (LKA), which assembles a linguistic knowledge base of real and fake indicators, and attention-based vision-language mutualism (AVM), which aligns multimodal features to enhance detection precision. Extensive experiments across 25 popular deepfake datasets demonstrate that LEDNet outperforms state-of-the-art methods, achieving superior accuracy and generalization across diverse deepfake generation techniques. This work sets a new benchmark in multimodal deepfake detection, advancing the application of foundation models in this domain.

Keywords computer vision, multimodal foundation model, deepfake detection

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1 Introduction

With the rapid advancement of deep learning-based generation technology [1–5], artificial intelligence (AI)-generated images are increasingly prevalent on social media platforms such as Twitter and WeChat. While these images enhance creativity and entertainment, they also pose significant safety risks. The ease of creating highly realistic images has raised concerns about misinformation, privacy, and security. Deepfakes, for example, can be used to disseminate fake information, leading to the spread of fake news that misleads the public. Moreover, these images can be exploited for malicious purposes, including identity theft, fraud, and cyberbullying, thus amplifying the potential for harm across social media platforms [6–10].

Researchers have made considerable efforts to mitigate these negative impacts through various approaches [10–13]. However, the generalization of deepfake detection models remains challenging, particularly due to the unknown origins of the generated images. This challenge is further compounded by the variety of generative models used to create these images, such as generative adversarial networks (GANs), diffusion models, and others. As generative models have emerged, this problem has become even more severe. Refs. [14, 15] have shown that state-of-the-art (SOTA) GAN detectors often fail to reliably distinguish real images from those generated by diffusion models, which makes challenges for generalization of deepfake detection.

In response, a significant amount of research [16, 17] initially focused on enhancing the generalization capability of deepfake detectors, particularly when training and test datasets differ in image styles and generative methods. However, these approaches remain limited and non-competitive, primarily due to

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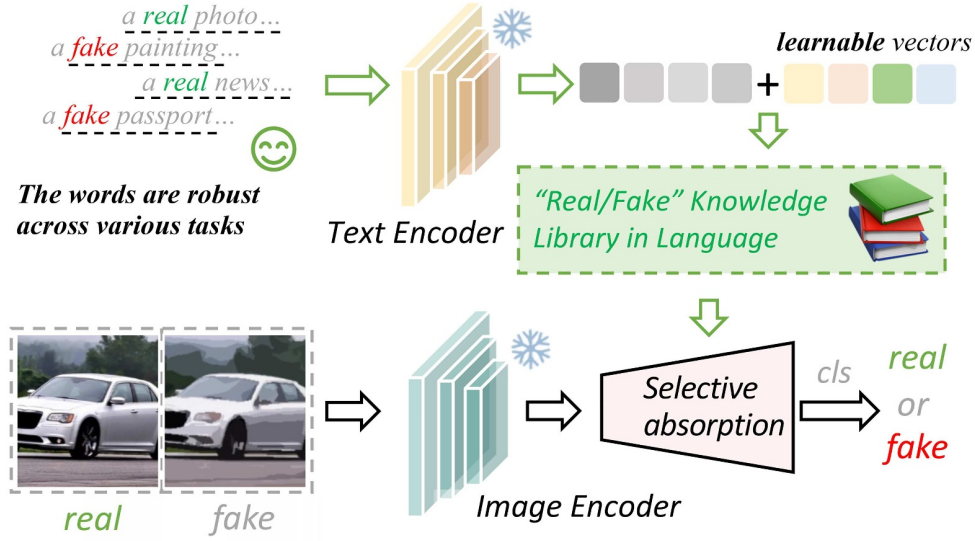


Figure 1 (Color online) Illustration of the proposed LEDNet. In language, the concepts of “real” and “fake” are highly versatile and robust across various tasks. Inspired by this, LEDNet incorporates language information, rich in semantics, to enhance the robustness of detecting forged images.

their reliance on specific artifacts and datasets. Recent efforts to enhance generalization also involve refining detection algorithms [18,19], augmenting datasets with more diverse images [20,21], and developing pre-trained models [13]. Despite these efforts, a significant gap persists: the absence of source-invariant representation leveraged from the generator pipeline. This deficiency leads to poor performance in detecting deepfake images generated from unknown technologies in the open-world scenario.

As illustrated in Figure 1, in this work, we introduce the language-enhanced deepfake detection network (LEDNet), which leverages linguistic descriptions of real and fake as auxiliary information for image identification. In natural language, the words “real” and “fake” can describe images generated by different models, as well as domains with significant differences, such as whether a woolen sweater is made entirely of wool. The effectiveness of these two simple words is robust across a range of complex and diverse tasks, inspiring us to enhance the robustness of forgery detection by incorporating linguistic information. Furthermore, some existing studies [22,23] have confirmed that automated prompt engineering, specifically for pre-trained vision-language models, outperforms manually designed language expressions in capturing semantic information. Therefore, we aim to enhance the accuracy and generalization of forgery detection models by identifying learnable vectors that effectively represent real and fake expressions in language.

In our approach, we use the contrastive language-image pre-training (CLIP) encoder as the backbone, leveraging its vast pre-trained knowledge base, which includes information about “real” and “fake” content. This pre-existing knowledge significantly enhances the deepfake detection process. While concerns have been raised about potential biases or limitations in the backbone model, we believe this is beyond the scope of our current study. Addressing such biases would require retraining or fine-tuning the CLIP model, which would deviate from our research focus. Additionally, many studies in the field also utilize pre-trained models from publicly available sources and assume them to be reliable feature extractors without considering biases as a core concern. We view the CLIP model’s pre-trained capabilities as a strong foundation for feature extraction, helping us to focus on the multimodal fusion and generalization aspects of deepfake detection.

The proposed LEDNet comprises two core modules: language-guided knowledge aggregation (LKA) and attention-based vision-language mutualism (AVM). The LKA module learns semantic information of real and fake, guided by manually designed expressions, and represents it as learnable feature vectors. Through interactive optimization with the image classification branch, LKA constructs a knowledge library in the linguistic domain, capturing the real and fake expressions of generated images. Additionally, the AVM module employs an attention mechanism that allows the image classification branch and the language knowledge library established by LKA to interact. This enables the model to selectively absorb relevant knowledge from the language knowledge library, supplementing the vision feature information of the input image. One of the key innovations of our work is the introduction of an un-trained-backbone mechanism. In this approach, we leverage a pre-trained vision-language model (CLIP) for feature extrac-

tion, without the need to retrain the model. This mechanism enables LEDNet to exploit both visual and linguistic information efficiently, improving generalization across diverse generative techniques.

To comprehensively evaluate the accuracy and generalization of the proposed model, we conducted extensive experiments on 25 well-known datasets. For accuracy, we trained and tested the model using images generated by the same method, such as ProGAN. To assess generalization, we trained the model on images generated by GAN-based methods and tested it on images produced by diffusion models. The final results from these experiments revealed that the proposed LEDNet not only achieves superior accuracy when trained and tested on images from the same generation method but also demonstrates exceptional generalization abilities when applied to images from different generative models. Thus, our contributions are mainly summarized into three different points below.

- To address the generalization problem in the deepfake detection task, we apply multi-modal technologies to the generalized deepfake detection task using an un-trained-backbone mechanism. This allows LEDNet to integrate powerful pre-trained vision-language models without additional retraining.
- We introduce two novel components of the proposed LEDNet: the LKA and AVM, which enable the fusion of visual and linguistic features for enhanced deepfake detection.
- We conduct extensive experiments on 25 well-established deepfake datasets, demonstrating that the proposed LEDNet outperforms state-of-the-art methods in both detection accuracy and generalization capabilities across different deepfake generation models.

2 Related work

Here, we engage in an in-depth exploration and comprehensive survey regarding the detection of AI-generated images as well as the generalization of deepfake detection.

2.1 AI-generated image detection

In recent years, many advanced methods have been proposed to handle the disastrous consequences caused by deepfakes. These studies' initial motivation stemmed from various perspectives. Ref. [11] firstly found that a standard image classifier can detect deepfake by careful pre- and post-processing and data augmentation. The work in [24] focused on efficiently utilizing both local and global image information. Refs. [25, 26] initially analyzed deepfake images in the frequency domain, and they found that the up-sampling operations lead to severe artifacts that can be easily identified. Therefore, a series of studies [27–29] exploited such artifacts to distinguish synthetic images from authentic ones. Similarly, Ref. [13] presented an artifact as an image gradient, thereby proposing a novel detection framework. Since research on multi-modality has achieved astonishing progress, particularly CLIP [30], Refs. [31, 32] introduced additional text information to assist the model in learning intricate artifacts, which shows the powerful potential of the vision-text model.

2.2 Generalization of deepfake detection

Despite those deepfake detection methods acquiring remarkable results on specific datasets, they gradually demonstrate their inability to address unprecedented AI-generated images, if they only rely on outdated and incomplete fingerprints. Refs. [33, 34] proved that the regularity of distribution in frequency can be easily erased, posing threats to algorithms that strongly depend on frequency artifacts. Refs. [14, 15] engaged a similar discussion regarding whether current detectors, developed for GAN-generated images, have superior performance on diffusion model (DM)-generated images. Their experiments exhibit that such a task is extremely difficult for those detectors. Given that current detectors and methods exhibit limited adaptability and generalization capability to rapidly evolving AI-generated images, numerous studies have dedicated academic endeavors and efforts towards tackling this arduous challenge. Ref. [16] simulated frequency artifacts to enable the detector to distinguish a deepfake of unknown source. Ref. [35] abandoned reliance on frequency artifacts and utilized multi-stage reconstruction loss to guide the detector. Ref. [36] addressed a novel method generated through the application of incremental learning. Conversely, Refs. [37, 38] conducted a thorough analysis to identify which features are the most crucial and common artifacts in synthetic images. Ref. [17] avoided including deepfake in the training dataset, and instead only used authentic images, employing up-sampling to mimic fake ones. Refs. [39, 40] respectively

adopted unsupervised aggregation and domain adaptation to enhance the generalization capability of the detector.

3 Method

In this section, we elaborate on the proposed LEDNet. First, we formalize the definition of generalized deepfake detection. Subsequently, we provide a comprehensive overview of the LEDNet framework. Finally, we introduce two core modules: LKA and AVM.

3.1 Problem definition

The main objective of generalized deepfake detection is to develop a forgery detector that can accurately identify deepfake images generated by unseen generative models. More concretely, this procedure can be formulated as follows. Given a set of image datasets S from N generators,

$$S = \{S_1, \dots, S_i, \dots, S_N\}, \quad (1)$$

S_i represents the i -th sub-dataset with fake images from the i -th generator and corresponding real images,

$$S_i = \{(x_1, y_1), \dots, (x_j, y_j), \dots, (x_K, y_K)\}, \quad (2)$$

where (x_j, y_j) denotes the j -th image and the corresponding category label (e.g., 0 for real and 1 for fake). Therefore, we solely train LEDNet with samples in S_i by minimizing the cross-entropy (CE) loss between the predicted and labels y_j . The formula is shown as follows:

$$\arg \min_{\theta} \sum_{j=1}^K \text{CE}(\text{LEDNet}(x_j; \theta), y_j), \quad (3)$$

where θ denotes the learnable parameters of LEDNet. During testing, we adopt the other sub-datasets $S - S_i$ to evaluate the cross-generator performance. As the testing samples are unseen in the training procedure, better performance represents better generalizability of deepfake detection.

3.2 Framework overview

In the model, we employ CLIP [30] as the backbone for feature extraction. CLIP has been pre-trained on vast amounts of text-image data and has developed a strong understanding of visual and linguistic associations, which we leverage to detect deepfake content. The pre-trained knowledge of “real” and “fake” concepts embedded within CLIP’s model significantly enhances the performance of our deepfake detection framework. While some may raise concerns about potential biases or limitations in the CLIP model itself, we believe this issue is outside the scope of our research. Our focus is on utilizing CLIP’s pre-trained capabilities for feature extraction rather than retraining or fine-tuning the model. Addressing any biases inherent in CLIP would require additional retraining or fine-tuning, which would shift the focus of our study from multi-modal integration and deepfake generalization to model reconfiguration.

As illustrated in Figure 2, LEDNet consists of two pre-trained encoders for extracting image and text features, which are from the CLIP model and frozen during training, as well as the proposed LKA to construct a linguistic knowledge library by gathering discriminative information of real and fake expressions and AVM to selectively absorb relevant knowledge from both vision and language domains by aggregating semantic-related features based on the context of input images. In LKA, we explore the concept of using shared learnable prompt vectors across different semantic texts, including those with opposing meanings, during the text fine-tuning process. Specifically, we investigated whether distinct semantic descriptions, such as “a real image captured by a camera” and “an AI-generated image”, should share the same prompt vector. Our hypothesis was that using a shared prompt vector could help the model generalize better across semantically related but distinct phrases. Finally, equipped with the proposed LKA and AVM, LEDNet is able to directly predict the real/fake classification results with a classifier. Through this attention-based visual-linguistic information interaction, LEDNet effectively avoids the computational cost associated with retraining the two encoders via contrastive learning in traditional multimodal learning tasks, which improves the inference efficiency of the model.

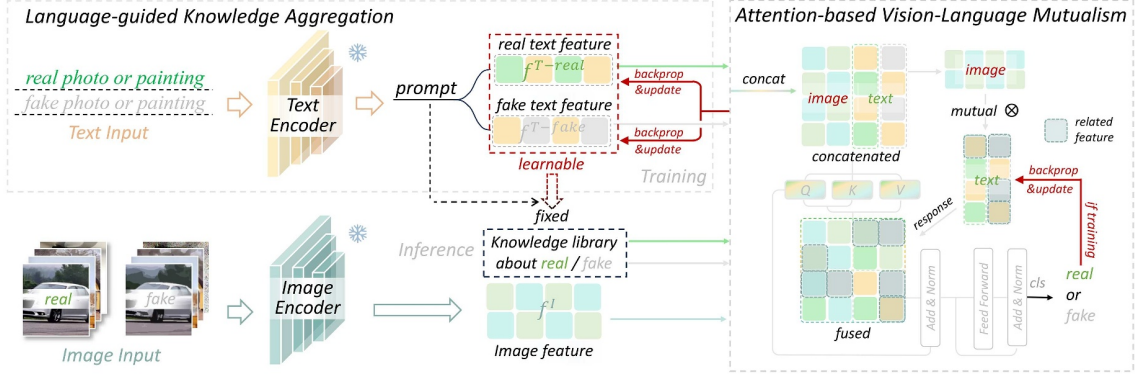


Figure 2 (Color online) Architecture of the proposed LEDNet, comprising two core modules: language-guided knowledge aggregation (LKA) and attention-based vision-language mutualism (AVM). LKA constructs a linguistic knowledge library that captures real and fake expressions of authentic and AI-generated images, while AVM selectively absorbs relevant knowledge from this library. It should be noted that the parameters of both the image and text encoders are frozen to reduce computational costs.

Unlike the famous Ojha [10] that only adopts the pre-trained CLIP-ViT (vision transformer) image encoder [30,41], LEDNet exploits the text encoder [42] to extract the “real” and “fake” descriptors within the linguistic domain to formulate a knowledge library with comprehensive understanding of deepfakes. Given an image $x \in \mathbb{R}^{3 \times H \times W}$, with height H and width W , LEDNet first converts it into D -dimensional image features $f^I \in \mathbb{R}^{(1+N) \times D}$ with the frozen image encoder. Here, 1 is the image classification (CLS) token and $N = \frac{HW}{P^2}$ denotes the image patch tokens with a patch size of P . Meanwhile, the frozen text encoder takes the language text descriptors t and drives the corresponding text embeddings $f^T \in \mathbb{R}^{M \times D}$, where M is the number of descriptors for constructing the knowledge library about real and fake.

3.3 Language-guided knowledge aggregation

Previous deepfake detection methods, such as those discussed by [10], primarily focus on the image domain to identify the forgery traces, thereby overlooking the valuable semantic-related information within the linguistic domain, which limits the generalization ability across unknown generative methods. To address this limitation, the proposed LKA module utilizes “real” and “fake” descriptors in natural language to construct a linguistic knowledge library with the aid of the frozen CLIP text encoder. Specifically, we manually compile attribute descriptions for general real and fake images, such as “real photo or painting” for real images and “fake photo or painting” for fake images. Instead of fine-tuning the text encoder to adjust descriptive text representations of real and fake knowledge, we freeze the CLIP text encoder’s original parameters and introduce learnable prompts p^T for f^T , and thus the formula is shown as follows:

$$\hat{f}_{(i)}^T = f_{(i)}^T + p^T, \text{ for } i = 1, \dots, M. \quad (4)$$

This approach avoids the computation cost of training a large number of parameters, preserving the modeling capability of the CLIP while enabling the model to differentiate between real and fake images by updating the learnable prompts p^T . During the inference, the learned linguistic knowledge library will be fixed, avoiding extra encoding costs and saving the training time.

3.4 Attention-based vision-language mutualism

After obtaining the descriptive representations from the linguistic knowledge library, a critical challenge emerges in effectively integrating these representations with the visual features derived from the image encoder. Traditional visual-language multimodal methodologies mainly rely on contrastive learning to align features across different modalities. However, these approaches require retraining both the image and language encoders to achieve this alignment, which may disrupt the well-generalized representations of the original CLIP model and lead to substantial computational costs.

To address this issue, we propose the AVM module. This module leverages attention mechanisms to selectively extract semantic-related features from the linguistic domain, tailored to the specific context of each image. By doing so, it utilizes the descriptive information of real and fake within the linguistic domain as auxiliary information, thus avoiding the computation cost for retraining the large number of parameters of the image and text encoder networks.

Algorithm 1 Procedure of LEDNet.

Input: The text fake descriptors t and image dataset $S_i = \{(x_1, y_1), \dots, (x_j, y_j), \dots, (x_K, y_K)\}$;
Output: The total cross-entropy loss L_{CE} and classification results $\{\hat{P}(y_1), \dots, \hat{P}(y_K)\}$;

- 1: **for** all $j = 1, 2, \dots, K$ **do**
- 2: Encode x_j and t into image and text features f^I and f^T with the frozen image encoder from the CLIP;
- 3: **for** all $i = 1, 2, \dots, M$ **do**
- 4: $\hat{f}_{(i)}^T = f_{(i)}^T + p^T$;
- 5: **end for**
- 6: Calculate W based on (5) and (6);
- 7: Calculate f_{forg} based on (7);
- 8: Calculate $\hat{P}(y_j)$ based on (8);
- 9: Calculate the loss based on (9) and add to L_{CE} ;
- 10: **end for**
- 11: Obtain the cross-entropy loss L_{CE} and classification results $\{\hat{P}(y_1), \dots, \hat{P}(y_K)\}$.

Specifically, given the encoded image features $f^I \in \mathbb{R}^{(1+N) \times D}$ and the forgery knowledge library $\hat{f}^T \in \mathbb{R}^{M \times D}$, we first employ a projection head to map all features into a shared forgery space. This step is crucial to mitigate the gap between the image and text modalities, facilitating more effective integration and utilization of the multimodal data.

$$\begin{aligned}\bar{f}^I &= \text{proj}(f^I), \\ \bar{f}^T &= \text{proj}(\hat{f}^T),\end{aligned}\tag{5}$$

where $\text{proj}(\cdot)$ is the shared projecting head. Then, to represent the specific context of each image, we adopt the image CLS token $\bar{f}_{(1)}^I$, as it has already gathered the global image features from the image encoder. Therefore, we have a formula which is shown as follows:

$$W = [\bar{f}^I, \bar{f}^T] \cdot (\bar{f}_{(1)}^I)^T,\tag{6}$$

where $[\cdot]$ is the concatenate operation along the channel dimension and $W \in \mathbb{R}^{(1+N+M) \times 1}$ gives the similarity representing the intensity between image context and encoded features from both visual and language domains. Next, we obtain semantic-related features via a selective gathering process, as follows:

$$f_{\text{forg}} = \text{softmax}(W) \cdot [\bar{f}^I, \bar{f}^T] + \bar{f}_{(1)}^I,\tag{7}$$

where $f_{\text{forg}} \in \mathbb{R}^{1 \times D}$ is the semantic-related feature. Finally, we can derive the prediction by applying a classification head ($\text{cls}(\cdot)$) on f_{forg} and further obtain the loss via the CE function:

$$\hat{P}(y) = \text{cls}(f_{\text{forg}}),\tag{8}$$

$$\text{CE} = -y \cdot \log \hat{P}(y) - (1 - y) \cdot \log(1 - \hat{P}(y)).\tag{9}$$

By considering (3), we obtain the final optimization function for training our LEDNet, as follows:

$$\arg \min_{\theta} \sum_{j=1}^K \text{CE}(\hat{P}(y_j), y_j).\tag{10}$$

The entire procedure of LEDNet is illustrated in Algorithm 1. Importantly, this study emphasizes the benefits of utilizing semantic-related information within the language domain and the vision-language mutualism in forgery detection, under the limited amount of training data. Following this, we conduct comprehensive evaluations in challenging real-world scenarios (primary and advanced generalization), encompassing a diverse set of 25 popular deepfake datasets.

3.5 Computational efficiency discussion

While the proposed model is relatively complex, we have ensured that its computational cost remains within acceptable bounds, making it suitable for practical deployment. The model was trained on a single NVIDIA 3090 GPU, which provided sufficient processing power to handle the computational requirements of the model. The high computational efficiency of the 3090 allowed us to train the model effectively without encountering performance bottlenecks, even with the complexity of the deepfake detection task.

This ensures that, despite the intricate architecture, the training process remains feasible for real-world applications, where time and resource constraints are common.

In addition to the manageable computational cost, the complexity of the model is balanced by the data requirements for training. Specifically, we trained our model on the ProGAN dataset, utilizing only four classes of images for training. This choice contrasts with traditional methods, which often rely on large, multi-class datasets (e.g., 20 or more classes) for training deepfake detection models. Our results demonstrate that the performance of our model does not degrade despite the smaller training set, allowing us to achieve state-of-the-art results with a significantly reduced dataset size.

This reduction in dataset size offers a significant advantage in terms of training efficiency. By focusing on fewer classes, we were able to achieve high detection accuracy and generalization without the need for extensive data augmentation or the computational overhead of working with larger datasets. This not only makes our method more efficient but also demonstrates that the model can achieve high performance without relying on massive amounts of training data, which is often a limitation in traditional deepfake detection models. Furthermore, our approach leverages the power of pre-trained models, such as CLIP, for feature extraction, which helps to further reduce training time and data requirements. Since the CLIP model has already been pre-trained on a large-scale dataset of text-image pairs, we avoid the need to retrain the model from scratch. Instead, we use the pre-trained embeddings to extract features, allowing us to focus on enhancing the multimodal fusion and deepfake detection capabilities of the model rather than expending resources on training large-scale visual encoders. This strategy significantly reduces the computational cost while maintaining the high accuracy and robustness of the model.

4 Experiments

In this section, we provide an intricate overview of the comprehensive experiments designed to thoroughly evaluate the effectiveness of the proposed LEDNet. We first introduce the experimental settings, including the dataset and implementation details. Then, we provide a detailed account of the preliminary experiment and various grouping experiments, which indicate the good performance of our method.

4.1 Settings

Datasets. To ensure a consistent basis for comparison, we use the training set from ForenSynths to train the detectors, following the approach used in previous baselines [11, 19, 21]. This training set includes 20 distinct categories, each with 18000 synthetic images generated using ProGAN [43], alongside an equal number of real images sourced from the LSUN dataset [44]. For evaluation, we utilized a comprehensive collection of 25 commonly used datasets. The first eight datasets are derived from ForenSynths [11], featuring images generated by eight distinct GAN-based models. The additional nine datasets come from GANGen-Detection [45], comprising images generated by nine different GANs. The remaining eight datasets are from DiffusionForensics [46], with the images generated using eight diffusion methods.

Implementation details. We use the Adam [47] as the optimizer with a learning rate of 1×10^{-4} and a batch size of 256. Our method is implemented using the PyTorch framework on Nvidia GeForce RTX 4090 GPU. We adhere to the evaluation metrics used in the baselines [10, 19, 21], which include the average precision score (A.P.) and the accuracy (Acc.).

4.2 Primary generalization evaluation

To evaluate the primary generalization ability of various models, we select images generated from one GAN model to train and images from other GAN-family generated models to test. Following the classic setting, we utilize the training data, a 4-class set (“car”, “cat”, “chair”, and “horse”) from the ForenSynths dataset. And the trained models are evaluated on two distinct datasets: ForenSynths and GANGen-Detection. The results are presented in Tables 1 [10–13, 18, 19, 21, 25, 33, 48, 49] and 2 [10–13, 18, 25, 33, 48, 49], comparing LEDNet with several SOTA methods across various GAN-based generative models.

Performance on ForenSynths dataset. As shown in Table 1, LEDNet consistently outperforms existing SOTA models across a range of generative models, including ProGAN, StyleGAN, and CycleGAN. Notably, LEDNet achieved the highest mean accuracy (90.8%) and average precision (99.0%), demonstrating its robust performance in detecting deepfakes generated by different GAN architectures. For instance, LEDNet outperformed models like LGrad and Ojha in specific datasets, such as BigGAN

Table 1 Results of the “primary generalization” evaluation. All the models are trained on a 4-class setting (ProGAN: “car”, “cat”, “chair”, and “horse”) and evaluated on the ForenSynths dataset (GAN-family generated models). The accuracy (Acc.) and precision (A.P.) are the evaluation metrics. The term “Mean” refers to the average performance across these testing sets. The best results are in bold. The second best results are underlined.

Method	ProGAN		StyleGAN		StyleGAN2		BigGAN		CycleGAN		StarGAN		GauGAN		Deepfake		Mean	
	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.
CNNDetection [11]	91.4	99.4	63.8	91.4	76.4	97.5	52.9	73.3	72.7	88.6	63.8	90.8	63.9	92.2	51.7	62.3	67.1	86.9
Frank [25]	90.3	85.2	74.5	72.0	73.1	71.4	88.7	86.0	75.5	71.2	99.5	99.5	69.2	77.4	60.7	49.1	78.9	76.5
Durall [33]	81.1	74.4	54.4	52.6	66.8	62.0	60.1	56.3	69.0	64.0	98.1	98.1	61.9	57.4	50.2	50.0	67.7	64.4
Patchfor [48]	97.8	100.0	82.6	93.1	83.6	98.5	64.7	69.5	74.5	87.2	100.0	100.0	57.2	55.4	85.0	93.2	80.7	87.1
F3Net [18]	99.4	100.0	92.6	99.7	88.0	99.8	65.3	69.9	76.4	84.3	100.0	100.0	58.1	56.7	63.5	78.8	80.4	86.2
SelfBland [12]	58.8	65.2	50.1	47.7	48.6	47.4	51.1	51.9	59.2	65.3	74.5	89.2	59.2	65.5	93.8	99.3	61.9	66.4
GANDetection [49]	82.7	95.1	74.4	92.9	69.9	87.9	76.3	89.9	85.2	95.5	68.8	99.7	61.4	75.8	60.0	83.9	72.3	90.1
BiHPF [19]	90.7	86.2	76.9	75.1	76.2	74.7	84.9	81.7	81.9	78.9	94.4	94.4	69.5	78.1	54.4	54.6	78.6	77.9
FrePGAN [21]	99.0	99.9	80.7	89.6	84.1	98.6	69.2	71.1	71.1	74.4	99.9	100.0	60.3	71.7	70.9	91.9	79.4	87.2
LGrad [13]	99.9	100.0	94.8	99.9	96.0	99.9	82.9	90.7	85.3	94.0	99.6	100.0	72.4	79.3	58.0	67.9	86.1	91.5
Ojha [10]	99.7	100.0	89.0	98.7	83.9	98.4	90.5	99.1	87.9	99.8	91.4	100.0	89.9	100.0	80.2	90.2	<u>89.1</u>	<u>98.3</u>
LEDNet (ours)	99.7	100.0	89.2	99.2	83.3	99.2	97.3	99.7	97.5	100.0	99.7	100.0	98.3	100.0	59.5	93.3	90.8	99.0

Table 2 Results of the “primary generalization” evaluation. All the models are trained on a 4-class setting (ProGAN: “car”, “cat”, “chair”, and “horse”) and evaluated on the GANGen-Detection dataset (GAN-family generated models). The best results are in bold. The second best results are underlined.

Method	InfoGAN		CraGAN		MMDGAN		AttGAN		BEGAN		RelGAN		S3GAN		SNGAN		STGAN		Mean	
	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.
CNNDetection [11]	71.1	94.7	81.5	97.5	72.9	94.4	51.1	83.7	50.2	44.9	53.3	82.1	55.2	66.1	62.7	90.4	63.0	92.7	62.3	82.9
Frank [25]	41.1	41.0	31.0	36.0	38.4	40.5	65.0	74.4	39.4	39.9	69.2	96.2	69.7	81.9	48.4	47.9	25.4	34.0	47.5	54.7
Durall [33]	50.1	51.7	60.9	67.2	59.5	65.5	39.9	38.2	48.2	30.9	80.0	88.2	87.3	97.0	54.8	58.9	62.1	72.5	60.3	63.3
Patchfor [48]	93.6	98.2	97.8	99.9	97.9	100.0	68.0	92.9	97.1	100.0	99.6	100.0	66.8	68.1	97.6	99.8	92.7	99.8	<u>90.1</u>	95.4
F3Net [18]	67.1	83.1	89.5	99.8	73.7	99.6	85.2	94.8	87.1	97.5	98.8	100.0	65.4	70.0	51.6	93.6	60.3	99.9	75.4	93.1
SelfBland [12]	79.0	82.5	75.1	82.4	68.6	74.0	63.1	66.1	56.4	59.0	73.6	77.8	53.2	53.9	61.6	65.0	61.2	66.7	65.8	69.7
GANDetection [49]	67.6	92.4	67.8	99.7	67.7	99.3	57.4	75.1	67.9	100.0	60.9	86.2	69.6	83.5	66.7	90.6	69.6	97.2	66.1	91.6
LGrad [13]	71.1	82.0	50.3	54.0	57.5	67.3	68.6	93.8	69.9	89.2	89.1	99.1	78.5	86.0	78.0	87.4	54.8	68.0	68.6	80.8
Ojha [10]	77.6	98.9	77.6	99.8	77.6	99.7	78.5	98.3	72.0	98.9	78.2	98.7	85.2	98.1	77.6	98.7	74.2	97.8	77.6	<u>98.8</u>
LEDNet (ours)	98.2	100.0	98.2	100.0	98.2	100.0	98.2	99.7	93.9	100.0	98.4	99.8	96.6	99.5	98.0	99.9	93.2	99.3	97.1	99.9

(Acc. 97.3%, A.P. 99.7%) and CycleGAN (Acc. 97.5%, A.P. 100.0%), indicating its superior ability of primary generalization across varying GAN-based generative models.

Performance on GANGen-Detection dataset. The results presented in Table 2, further validate the generalization strength of LEDNet. The model achieved an impressive mean accuracy of 97.1% and an average precision of 99.9%, surpassing all other compared methods. LEDNet’s performance is particularly noteworthy in the detection of images generated by challenging models like InfoGAN, CraGAN, and MMDGAN, where it consistently maintained high accuracy and precision, highlighting its capability to handle other diverse and complex deepfake detection tasks.

Across both datasets, LEDNet exhibits a significant improvement over traditional methods such as CNNDetection, Frank, and Durall, which show variable performance across different generative models. While some models like Ojha and Patchfor achieve high precision in certain scenarios, LEDNet’s balanced performance across all tested categories underscores its effectiveness in both accuracy and generalization, making it a versatile and reliable choice for deepfake detection. The comprehensive experimental results demonstrate that LEDNet not only achieves superior accuracy in individual categories but also excels in generalization across a broad range of GAN-based models. This makes LEDNet a powerful tool for robust deepfake detection, capable of adapting to various generative methods without the need for retraining image and text encoders, thereby streamlining the detection process while maintaining high performance.

4.3 Advanced generalization evaluation

To assess the advanced generalization capabilities of the proposed LEDNet model, we conducted experiments for the “advanced generalization” evaluation, wherein all models were trained on a 4-class setting

Table 3 Results of the “advanced generalization” evaluation. All the models are trained on a 4-class setting (ProGAN: “car”, “cat”, “chair”, and “horse”) and evaluated on the DiffusionForensics dataset (Diffusion-based generated models). The best results are in bold. The second best results are underlined.

Method	Glide_100_10		Glide_100_27		Glide_50_27		ADM		DALLE		LDM_100		LDM_200		LDM_200_cfg		Mean	
	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.
CNNDetection [11]	54.9	66.6	51.8	61.3	53.3	72.9	53.0	71.3	54.2	76.0	51.9	63.7	52.0	64.5	51.6	63.1	52.8	67.4
Frank [25]	53.4	52.5	57.0	62.5	53.6	44.3	50.4	40.8	52.0	42.3	56.6	51.3	56.4	50.9	56.5	52.1	54.5	49.6
Durall [33]	40.6	42.3	55.9	58.0	54.9	52.3	48.9	46.9	51.7	49.9	62.0	62.6	61.7	61.7	58.4	58.5	54.3	54.0
Patchfor [48]	74.2	81.4	79.8	99.1	87.3	99.7	82.8	99.1	84.9	98.8	95.8	99.8	95.6	99.9	94.0	99.8	86.8	97.2
F3Net [18]	69.2	70.8	71.6	79.9	88.3	95.4	87.0	94.5	88.5	95.4	74.1	84.0	73.4	83.3	80.7	89.1	79.1	86.5
SelfBland [12]	58.3	63.4	52.4	51.6	58.8	63.2	59.4	64.1	64.2	68.3	53.0	54.0	52.6	51.9	51.9	52.6	56.3	58.7
GANDetection [49]	49.6	49.0	67.2	83.0	51.2	52.6	51.1	51.9	51.7	53.5	54.7	65.8	54.9	65.9	53.8	58.9	54.3	60.1
LGrad [13]	86.6	100.0	88.5	97.3	89.4	94.9	87.4	93.2	90.7	95.1	94.8	99.2	94.2	99.1	95.9	99.2	90.9	<u>97.1</u>
Ojha [10]	89.5	96.8	90.1	97.0	90.7	97.2	91.1	97.4	75.7	85.1	90.5	97.0	90.2	97.1	77.3	88.6	86.9	94.5
LEDNet (ours)	90.9	98.1	90.2	98.1	91.7	98.4	73.4	93.0	92.5	98.0	93.3	98.9	93.2	98.8	77.5	92.7	<u>87.9</u>	<u>97.1</u>

(ProGAN: “car”, “cat”, “chair”, and “horse”) and tested on the DiffusionForensics dataset. This dataset comprises images generated by various diffusion-based models, representing a significant challenge for deepfake detection, and results are presented in Table 3 [10–13, 18, 25, 33, 48, 49].

In Table 3, LEDNet demonstrated strong performance across the majority of diffusion-based models, achieving an average accuracy of 87.9% and an impressive average precision of 97.1%. Notably, LEDNet excelled in detecting images generated by Glide_100_27 (Acc. 90.2%, A.P. 98.1%) and DALLE (Acc. 92.5%, A.P. 98.0%), showcasing its robust ability to generalize across different diffusion techniques. This indicates that LEDNet is highly effective in scenarios where generative methods diverge significantly from those seen during training. When compared to SOTA baselines, LEDNet performed competitively, particularly in terms of precision. While LGrad achieved a slightly higher average accuracy (90.9%) and precision (97.2%), LEDNet’s balanced performance across various diffusion models highlights its versatility. For example, LEDNet surpassed models like CNNDetection and Frank, which struggled with lower accuracy and precision, particularly on challenging models such as Glide_50_27 and ADM. Moreover, LEDNet outperformed the Ojha model in terms of precision on multiple datasets, despite Ojha’s strong performance in specific categories. This suggests that while Ojha may excel in certain tasks, LEDNet offers more consistent performance across a wider range of generative models, making it a reliable choice for unknown generative methods like diffusion models.

The experimental results from the “advanced generalization” evaluation of this section confirm that LEDNet effectively generalizes to unknown generative methods, particularly those based on diffusion processes. This is a critical capability for real-world applications where the diversity of generative models continues to expand. LEDNet’s high precision across different datasets reinforces its potential as a robust model for accurately detecting deepfakes, even in the face of advanced and varied generative techniques.

4.4 Ablation study

In this section, we perform a more in-depth analysis of the different settings in our experiment. The effect of separating the LKA and AVM modules on the results of the experiment was not discussed in our ablation experiments for the following reasons. In our model, the LKA module primarily extracts forged features at the semantic level, while the AVM module facilitates the fusion of text and image features through a self-attention mechanism. These two modules are complementary and indispensable, both playing a crucial role during inference and training. As a result, it is not feasible to separate them in ablation experiments without compromising the integrity of our approach.

4.4.1 Prompt setting

The experimental results presented in Figure 3 underscore the critical influence of prompt design on the accuracy of deepfake detection across eight distinct generative models. Notably, the learnable prompt we proposed (E) outperforms all other prompt types, achieving the highest mean accuracy of 86.1%. This superiority is attributed to the learnable nature of the prompt, which enables models to dynamically adjust and refine the language library during training. The results suggest that even subtle enhancements in prompt design can lead to significant improvements in model performance, particularly in differentiating

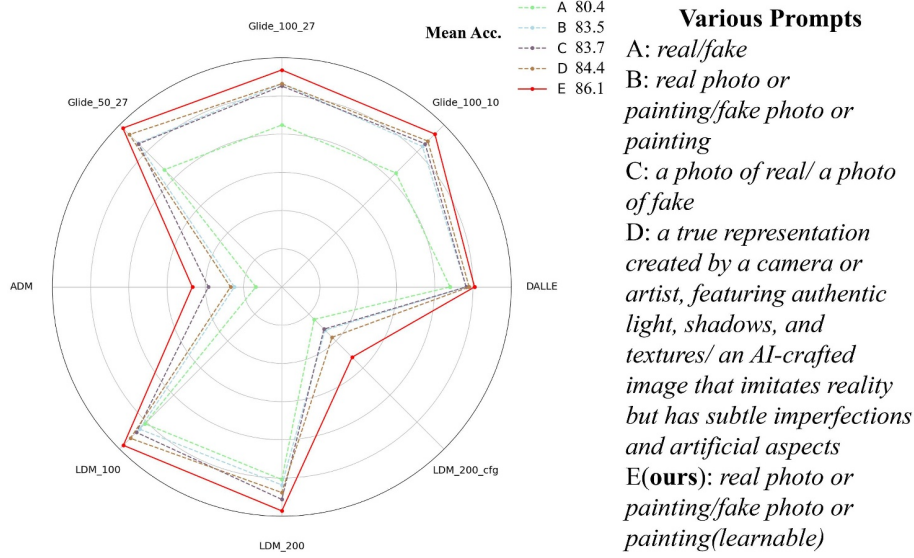


Figure 3 (Color online) Performance of different manual prompts.

Table 4 Performance comparison of different vision-language fusion strategies. The strategies are as follows: (A) addition with equal proportions, where visual and language features are combined with no weighting; (B) addition with weighted coefficients, allowing for the adjustment of importance; (C) concatenation, where features from both modalities are merged into a single; (D) attention-based strategy, which selectively emphasizes relevant features from both domains to optimize performance. The best results are in bold. The second best results are underlined.

Strategy	DALI		Glide_100_10		Glide_100_27		Glide_50_27		ADM		LDM_100		LDM_200		LDM_200_cfg		Mean	
	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.
(A)	76.2	95.1	73.0	96.4	70.4	95.4	73.6	96.6	55.3	83.3	75.0	95.6	74.2	95.5	57.7	83.8	71.4	94.0
(B)	77.5	96.8	74.0	97.7	71.4	96.9	76.2	98.0	56.1	85.6	76.5	97.4	75.9	97.3	60.0	89.5	73.0	96.2
(C)	81.9	96.3	82.4	97.9	79.2	97.2	84.4	98.2	59.0	88.8	81.5	97.2	81.2	97.1	62.1	87.4	<u>78.9</u>	<u>95.9</u>
(D) (ours)	85.2	94.9	88.3	96.4	88.4	96.5	89.4	96.9	71.7	91.5	89.3	96.9	89.3	96.8	73.0	86.9	86.1	95.1

between real and AI-generated images. Furthermore, while simpler prompts like prompt (A) provide a strong baseline, the more contextually rich descriptions in prompts (B) and (C) also exhibit improved accuracy, highlighting the importance of detail in prompt construction. However, overly complex prompts, as seen in prompt (D), may introduce ambiguity, slightly diminishing performance. These findings underscore the delicate balance between simplicity and detail in prompt design, with our proposed strategy achieving the optimal balance for maximizing model efficacy.

4.4.2 Vision-language fusion strategy

Table 4 presents the performance of different vision-language fusion strategies across various datasets. Notably, our proposed attention-based strategy (D) significantly outperforms the other methods, achieving the highest mean accuracy of 86.1% and an average precision (A.P.) of 95.1%. These results underscore the effectiveness of leveraging attention mechanisms to enhance the integration of visual and linguistic features, enabling more accurate and precise model predictions. In contrast, simpler fusion strategies, such as addition with equal proportion (A) and weighted coefficients (B), yield lower performance, with mean accuracies of 71.4% and 73.0%, respectively. The concatenation method (C) shows some improvement, attaining a mean accuracy of 78.9%, yet still falls short of the attention-based approach. These findings emphasize the superiority of attention mechanisms in capturing the relations between vision and language, leading to more robust performance across diverse datasets.

4.4.3 Dimension analysis of vision modality

To thoroughly evaluate the influence of the number of image feature channels on the model's performance, we conducted a series of experiments. In one such experiment, we focused on a simplified scenario where only the image CLS token was fused with text features, utilizing a self-attention mechanism. This approach allowed us to isolate and better understand the contribution of the CLS token in the context

Table 5 Results of the “primary generalization” evaluation. All models are trained on a 4-class setting (ProGAN: “car”, “cat”, “chair”, and “horse”) and evaluated on the ForenSynths dataset (GAN-family generated models). Evaluation metrics include accuracy (Acc.) and average precision (A.P.). “Mean” refers to the average performance across these testing sets. The models are defined as follows: pt using a single image feature channel fused with text features; pt2 using the first 128 image feature channels fused with text features; pt3 using the last 129 image feature channels fused with text features; pf using conventional methods to learn prompt word vectors at the input end; r-f using different prompt vectors for different text descriptions. The best results are in bold. The second best results are underlined.

Method	ProGAN		StyleGAN		StyleGAN2		BigGAN		CycleGAN		StarGAN		GauGAN		Deepfake		Mean	
	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.
LEDNet (ours)	99.7	100.0	89.2	99.2	83.3	99.2	97.3	99.7	97.5	100.0	99.7	100.0	98.3	100.0	59.5	93.3	90.8	99.0
pt	99.8	100.0	87.6	98.4	81.2	98.7	95.2	99.4	93.7	99.7	97.6	100.0	97.6	100.0	71.9	91.9	90.6	<u>98.5</u>
pt2	99.5	99.9	83.7	98.2	73.9	98.0	96.9	99.5	95.5	99.9	96.7	99.9	98.6	99.9	76.0	91.6	90.1	98.4
pt3	99.2	99.9	87.5	98.1	80.4	95.9	96.5	99.6	92.6	98.7	99.0	99.9	98.5	99.9	61.4	82.6	89.4	96.8
pf	99.4	100.0	85.3	98.4	78.4	98.5	96.7	99.5	97.8	99.9	93.4	99.9	98.2	100.0	77.9	89.9	<u>90.8</u>	98.3
r-f	99.1	100.0	79.8	97.5	68.7	97.8	95.2	99.2	97.6	99.9	99.4	100.0	98.8	100.0	61.2	91.5	87.5	98.2

Table 6 Results of the “primary generalization” evaluation. All models are trained on a 4-class setting (ProGAN: “car”, “cat”, “chair”, and “horse”) and evaluated on the GANGen-Detection dataset (GAN-family generated models). The best results are in bold. The second best results are underlined.

Method	InfoGAN		CraGAN		MMDGAN		AttGAN		BEGAN		RelGAN		S3GAN		SNGAN		STGAN		Mean	
	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.
LEDNet (ours)	98.2	100.0	98.2	100.0	98.2	100.0	98.2	99.7	93.9	100.0	98.4	99.8	96.6	99.5	98.0	99.9	93.2	99.3	97.1	99.9
pt	83.5	99.8	83.6	100.0	83.6	99.9	94.5	99.1	72.6	99.8	95.3	99.3	93.3	98.8	83.5	99.5	77.4	97.5	85.3	99.3
pt2	88.9	99.9	88.9	99.9	88.9	99.9	95.2	99.0	77.0	99.9	96.7	99.6	95.5	99.1	88.8	99.6	75.4	97.7	88.3	99.4
pt3	98.3	99.8	99.4	99.9	98.3	99.9	84.4	96.8	99.2	99.9	94.0	99.1	94.8	99.2	90.3	99.1	90.5	97.6	94.4	99.1
pf	79.7	99.8	79.7	100.0	79.6	99.9	89.4	98.3	65.4	99.8	89.5	98.7	95.9	99.4	79.6	99.2	66.3	97.5	80.6	99.2
r-f	97.3	99.8	97.4	99.9	97.4	99.9	94.7	99.0	92.6	99.8	96.4	99.5	93.2	99.0	96.6	99.5	88.5	97.7	<u>94.9</u>	<u>99.4</u>

Table 7 Results of the “advanced generalization” evaluation. All the models are trained on a 4-class setting (ProGAN: “car”, “cat”, “chair”, and “horse”) and evaluated on the DiffusionForensics dataset (diffusion-based generated models). The best results are in bold. The second best results are underlined.

Method	Glide_100_10		Glide_100_27		Glide_50_27		ADM		DALLE		LDM_100		LDM_200		LDM_200_cfg		Mean	
	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.	Acc.	A.P.
LEDNet (ours)	90.9	98.1	90.2	98.1	91.7	98.4	73.4	93.0	92.5	98.0	93.3	98.9	93.2	98.8	77.5	92.7	87.9	97.1
pt	86.8	96.9	86.8	96.8	87.9	97.3	71.4	92.6	87.5	96.7	89.9	97.6	89.5	97.7	70.9	89.1	<u>85.6</u>	96.0
pt2	86.4	97.0	85.2	96.9	87.4	97.6	67.1	91.4	86.8	96.9	90.8	98.2	90.1	97.9	69.9	89.1	85.2	96.3
pt3	76.3	94.1	77.6	93.9	78.9	94.9	62.5	90.9	85.5	95.9	89.4	98.2	89.9	98.2	73.4	91.5	81.6	95.3
pf	83.7	97.5	82.6	97.5	85.3	97.9	65.2	92.5	81.5	96.3	86.3	97.7	86.6	97.7	67.3	89.5	81.9	<u>96.3</u>
r-f	75.1	95.2	73.7	94.7	75.9	95.6	58.6	88.6	79.7	95.8	79.7	96.0	79.2	95.7	61.8	85.0	75.1	94.0

of image-text fusion, particularly when the number of image feature channels is minimized. Moreover, to explore the impact of the number of image feature channels on the experimental results, we also conducted experiments using only the first 128 dimensional image features and the last 129 dimensional image features separately (with a total of 257 image feature channels). The results, which provide insights into how this simplification impacts the performance, are detailed in Tables 5–7 below.

Tables 5–7 compare the outcomes of the different experimental setups, highlighting how variations in the number of image feature channels influence the model’s ability to accurately align and integrate visual and textual information. As shown in the “pt” row in Tables 5–7, the result reveals the impact of reducing the number of feature channels on the image side. Interestingly, when the number of image feature channels is reduced, the performance of the model in Tables 5 and 7 shows only a minimal decline. Specifically, the average accuracy (Acc.) in Table 5 decreases by just 0.2%, and in Table 7, the decrease is only 2.3%. These minor reductions indicate that LEDNet is capable of maintaining high performance even with fewer image feature channels in certain contexts. However, the scenario is different in Table 6, where the reduction in image feature channels results in a significant drop in performance. The average Acc. decreases by 11.8%, reflecting a notable decline in the model’s effectiveness, particularly across most datasets in Table 6 (with the exceptions of AttGAN, RelGAN, and S3GAN). This considerable reduction in accuracy suggests that in these cases, LEDNet’s ability to extract and leverage robust image features is heavily dependent on the availability of multiple image channels. At the same time, as shown in the

“pt2” and “pt3” rows in Tables 5–7, the reduction of image feature channels will overall lead to poorer performance of the model. For example, the average accuracy of “pt2” in Table 5 decreases by 0.7%, in Table 6, the decrease is 8.8%, and in Table 7, it also decreases by 2.7%.

The observed results underscore the importance of maintaining a sufficient number of image feature channels to ensure robust performance across diverse datasets, particularly when dealing with complex or varied data distributions. The results not only demonstrate the robustness of the LEDNet architecture under different conditions but also emphasize the critical role that the CLS token and the number of image feature channels play in achieving optimal performance. By presenting this detailed analysis, we aim to provide a comprehensive understanding of the underlying dynamics within LEDNet, ultimately contributing to a deeper appreciation of LEDNet’s capabilities and limitations in different deepfake tasks.

4.4.4 *Position and implementation types of learnable vectors*

In text-related tasks, the conventional methodology typically involves placing learnable prompt words at the input stage of the model, specifically at the input of the text encoder. This traditional approach relies on the premise that the model can effectively process and integrate these prompts from the very beginning of the text encoding process, thereby influencing the subsequent layers and ultimately guiding the model’s understanding and generation of text. However, in our work, we have adopted a novel and innovative approach that diverges from this established norm. Instead of positioning the learnable prompt words at the input, we focus on learning the output vector of the text encoder, effectively treating it as a specialized task involving a unique type of learnable prompt word. This output-based approach is designed to capture and refine the textual representations after they have been processed by the text encoder, allowing for a more nuanced and contextually aware integration of prompts into the proposed LEDNet’s decision-making process.

The proposed method not only challenges the conventional wisdom but also opens up new possibilities for enhancing model performance by leveraging the rich contextual information that is available at the output stage of the text encoder. By focusing on the output vector, we aim to refine and optimize the model’s ability to understand and generate text, particularly in complex scenarios where the nuances of context and semantics play a critical role. To thoroughly assess the effectiveness of LEDNet, we also conducted a series of experiments that involved the traditional method of learning prompt word vectors at the input stage. This comparison allowed us to directly evaluate the relative merits of each approach, providing a clear understanding of how they influence the model’s performance. The specific effects of the conventional input-based method are meticulously documented in Tables 5–7 below. It can be found from the “pf” row in Tables 5–7 that when we use traditional prompt word learning methods to fine-tune the prompt vector model at the input of the text encoder, the effect will also be significantly reduced. Compared with LEDNet (ours), the average Acc. in Table 5 is the same (Acc. 90.8%). However, training prompt word vectors at the input of the text encoder will greatly reduce the performance metrics of Tables 6 and 7. The average Acc. decreased by 16.5% in Table 6 and the average Acc. decreased by 6.0% in Table 7.

To investigate whether distinct semantic descriptions, such as “a real image captured by a camera” and “an AI-generated image”, should share the same prompt vector, we implemented a mechanism using different prompt vectors for different text descriptions (as shown in the “r-f” rows of Tables 5–7). The results indicate that this approach generally leads to a decrease in performance across Tables 5–7. Specifically, compared to our model (LEDNet), the average accuracy in Table 5 decreased by 3.3%, in Table 6 by 2.2%, and in Table 7 by 12.8%. Through a comparative analysis, we observed how performance metrics varied when using shared versus distinct prompt vectors. This analysis highlights not only the impact of our proposed method but also the trade-offs associated with each approach.

The analysis of these experimental results offers compelling evidence in support of our proposed method. By examining various performance metrics and scenarios, it becomes evident that our approach of learning the output vector as a type of learnable prompt word significantly outperforms the traditional input-based method. The data suggests that our model benefits from the additional context and refined representations available at the output stage, leading to superior performance in text-related tasks. This innovative strategy not only enhances the model’s accuracy and efficiency but also contributes to a deeper understanding of how prompt learning can be optimized within complex text processing systems, providing different insights in training prompt vectors.

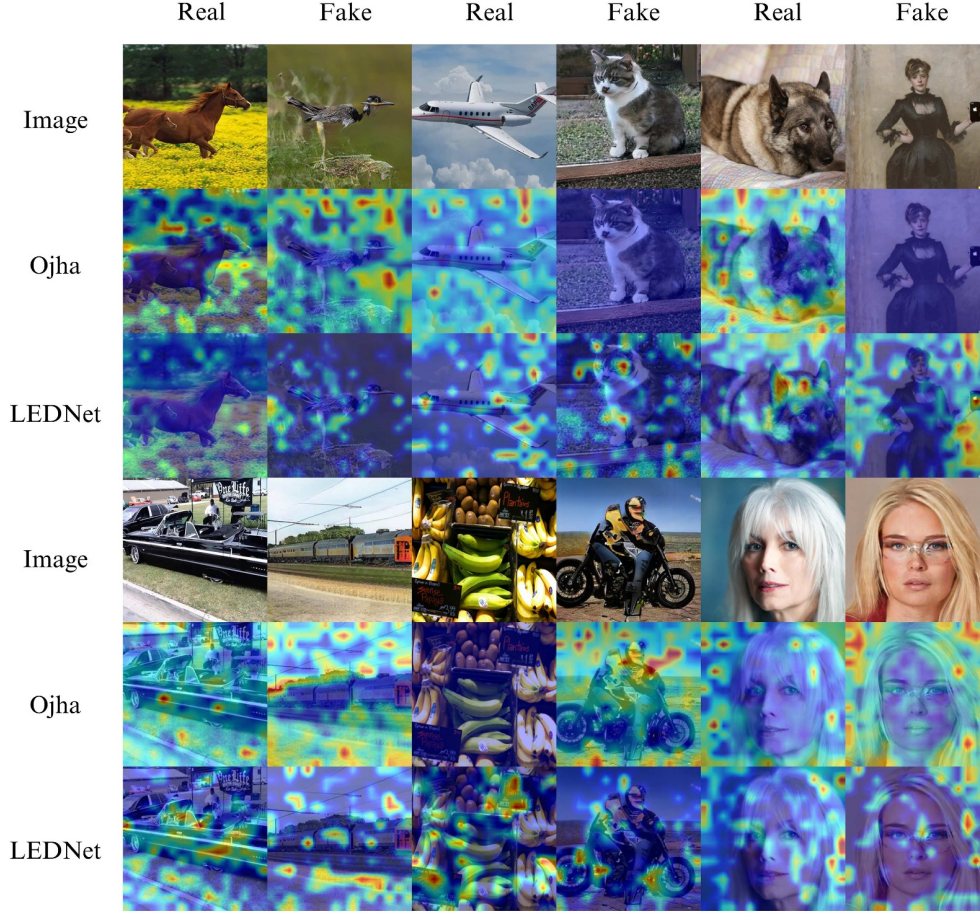


Figure 4 (Color online) Attention visualization comparative analysis.

4.5 Attention visualization

To effectively demonstrate that the proposed LEDNet focuses on image semantics through the integration of language information, we conducted an attention visualization experiment. This experiment, as vividly illustrated in Figure 4, aimed to provide a general understanding of how LEDNet operates in relation to language-integrated image analysis. We also compared LEDNet with the baseline model developed by Ojha, which stands as the latest work incorporating the CLIP image encoder. The result obtained from the experiment is truly remarkable. It clearly emphasizes that LEDNet integrates higher-level semantic understanding. This semantic understanding is significantly influenced by language cues, which act as guiding beacons in the decision-making process during deepfake detection. By leveraging these language cues, LEDNet is able to make more informed and accurate decisions, thereby enhancing its overall performance in identifying deepfakes that are generated by different models.

In the experiment, the attention maps generated by LEDNet revealed that the model has an innate ability to concentrate on specific regions of the image where semantic discrepancies may indicate a fake. For instance, when LEDNet processes a face image, it focuses intently on critical areas such as the eyes. Any subtle alterations in these areas can be a tell-tale sign of tampering, and LEDNet is able to zero in on these details with remarkable precision. Moreover, when handling a diverse range of images, including those of airplanes, cars, dogs, and other objects, our model demonstrates its prowess in identifying the most vulnerable areas. In the case of airplanes, it focuses more on the wings, which are areas that could potentially be modified in a deepfake. For cars, the car cockpits receive special attention, as these are key areas where forgeries might be attempted. Similarly, when it comes to dog images, the dog heads are the main focus, as they are areas where signs of manipulation are more likely to be found. This indicates that our model can effectively extract information that is highly conducive to forgery detection under the guidance of language modalities, which leads to strong generalization ability.

These findings strongly reinforce the initial hypothesis that LEDNet's integration of language informa-

tion empowers it to interpret images in a more context-aware manner. This enhanced context-awareness becomes particularly evident in complex scenarios. For example, in situations where the generative models used during training and testing differ, LEDNet's attention maps exhibit a consistent and remarkable focus. They focus on regions that align precisely with the learned linguistic descriptions of "real" and "fake". These visualizations serve as a powerful testament to the model's capability to generalize across different generative methods. By utilizing language as an additional semantic enhancement, LEDNet is able to overcome the challenges posed by variations in generative models and therefore significantly improve its deepfake detection performance in complex tasks.

5 Conclusion

In this paper, we introduced the LEDNet, which integrates linguistic information into deepfake detection to enhance the generalization capabilities. LEDNet is designed with two remarkable and highly effective modules, namely LKA and AVM. The LKA module is specifically engineered to extract and organize knowledge from language-based information, leveraging the rich semantic content within the language cues. It acts as a guiding force, directing the network to focus on relevant aspects within the potentially fake data, which plays an important role in deepfake detection. On the other hand, the AVM module plays a crucial role in integrating useful visual and textual information. By leveraging the LKA and AVM modules, LEDNet effectively merges language-based features with visual data, enabling robust detection performance across a wide range of generative models. Our experiments were conducted on 25 diverse and popular deepfake datasets. Thus, the results clearly demonstrate LEDNet's superior performance, particularly in achieving high accuracy within the consistent generative methods (primary generalization) and exceptional generalization across different models (advanced generalization). This indicates that even when faced with deepfakes produced by entirely different generative models during the testing phase compared to what it was trained on, LEDNet is still effective.

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